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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I016*

Team Name: *CODEHOLICS*

Domain: *Food Security, Agriculture Rural Transformation*

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Presentation Outline

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Problem Statement

Farming the traditional way often means guessing. Water and nutrients get used less efficiently than they should, and a sudden change in the weather can undo days of work. Farmers also face an awkward choice: automate everything and give up control, or keep doing it all by hand. Without a live view of what's happening in the field, manual irrigation tends to waste water and throw nutrients out of balance. Meanwhile, most automated systems take the decision out of the farmer's hands and have no safety net to bring a person back in when something goes wrong.

Problem Understanding

- [Who experiences this problem?]
  Farmers and farm workers, from small plots to large fields, who spend their days on heavy, repetitive manual work. Checking soil, opening valves, and walking the rows all take time and energy. A system that takes some of that load off, without taking away their say, makes a real difference to their

- [What causes the problem?]
  1. Soil data is missing: Without buried sensors, soil moisture and nutrients (NPK) stay invisible, so watering and fertilizing become guesswork.
  2. Irrigation that can't react: Fixed timers and manual switches can't respond to extreme heat or intense sunlight. Water comes on schedule, not on need.
  3. Tired people, large fields: In manual mode, one person covering acres can easily miss a dry patch or forget a pump, and the crop suffers unnoticed.

- [How does it affect users or stakeholders?]
  Economic Losses: Over-fertilization wastes money, while under-fertilization causes stunted crop growth and reduced market value.
  Resource Depletion: Excessive irrigation wastes precious groundwater and spikes electricity costs.

Existing Gaps & Challenges

- 1. High AC Power Grid Dependency: Existing automated systems rely on continuous grid power, making them unusable in remote fields subject to power outages.

- 2. Broad Surface Spray vs. Targeted Delivery: Standard setups spray water broadly over open fields, wasting resources and feeding weeds.

- [Main challenge that remains unsolved]
  1.Remote Grid Dependency
  Unsolved: Continuous AC power needed; fails during rural power outages.
  Our Solution: Off-grid solar power for 24/7 reliability.
  2. Water Fertilizer Waste
  Unsolved: Surface spraying loses water to evaporation and feeds weeds. Our Solution: Root-zone delivery via main-to-branch lines.
  3. Manual Mode "Blind Spot"
  Unsolved: Switching to manual disables all safety alerts. Our Solution: Background safety monitoring stays active in manual mode.

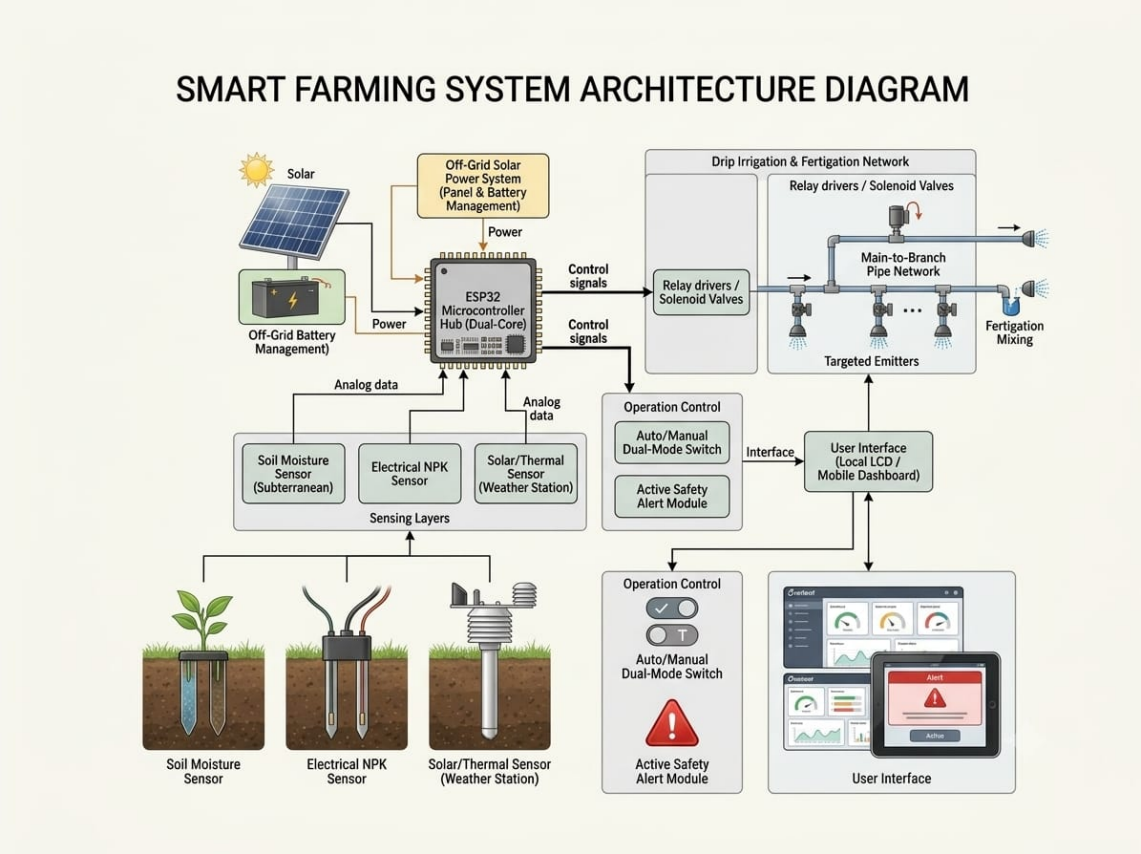
Our Proposed Solution

- [Core idea or product]
  An off-grid, ESP32-powered smart farming system integrating subterranean soil moisture and NPK nutrient monitoring with solar-responsive irrigation

- [How the solution works at a high level]
  It automates root-zone delivery via main-to-branch lines, while giving farmers a flexible dual-mode control interface backed by a 24/7 active safety-alert network.

- [Main value delivered to users]
  Maximizes crop yield and resource efficiency with zero grid dependency, eliminating water waste while keeping complete safety control in the farmer's hands.

Technical Approach



[Explain how the system components work together.]

Tech Stack

Frontend: [Add here]

Backend: [Add here]

Database: [Add here]

AI/ML, APIs, or Hardware: [Add if applicable]

Innovation & Uniqueness

- [What is new or different?]
  Integrates subterranean NPK and moisture sensing into a single off-grid hub with continuous background safety monitoring.

- [Why is this approach suited to the problem?]
  Solves grid instability and resource waste through localized ESP32 logic and targeted root-zone delivery.

- [What is the key differentiator?]
  Zero "manual mode" blind spots—background safety alerts stay active 24/7, even during manual control.

Feasibility & Scalability

- **Feasibility:** Uses low-cost hardware (ESP32, solar PV, basic NPK/moisture sensors). Built in three steps: sensor calibration, pipe integration, and firmware setup.
- **Scalability:** Modular design allows adding extra sensor nodes and valves via wireless mesh (ESP-NOW), scaling easily from small plots to large farms.
- **Risks:** Risk: Solar power drops or sensor fouling from harsh weather. Mitigation: Battery backup, protective sensor enclosures, and active manual overrides

Cost for Product (Optional)

- Development or hardware cost: 4,500 — ESP32, subterranean NPK moisture sensors, solar panel, battery, relays, and piping.
- Hosting or maintenance cost: 500/year — Near-zero cloud costs due to local-first processing; covers periodic battery sensor upkeep.
- Estimated total: 5,000 INR for prototype deployment.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Off-Grid Remote Farming (Rural Smallholder)
  A farmer with a plot in an area prone to daily power grid cuts relies on the system's solar power to maintain automated subterranean irrigation without manual power management

- Use case 2: Manual Mode Safety Intervention (Human Error Prevention)
  A field worker switches to manual irrigation and forgets to turn off a valve; background monitoring triggers an active safety alert before over-watering occurs.

- Use case 3: Targeted NPK Fertigation (Nutrient Optimization)
  During crop growth, subterranean sensors detect low nitrogen levels and automatically mix/deliver liquid NPK directly to the root zone via branch lines, preventing fertilizer waste

Impacts and Benefits

- [Benefit to the target users]
  Provides off-grid farmers with automated, 24/7 root-zone irrigation and NPK delivery, while preventing crop loss via continuous background safety monitoring

- [Social, economic, or environmental impact]
  Economic Environmental: Reduces water evaporation and fertilizer runoff, lowers input costs, and eliminates reliance on expensive AC grid power.

- [Expected measurable outcome]
  Achieves up to 30–40(percent) water savings, 20(percent) reduction in fertilizer waste, and 100(percent) off-grid operational uptime.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.4	Zero Hunger	Ensures sustainable food production systems through precision NPK fertigation and root-zone care.
SDG 6	Target 6.4	Clean Water and Sanitation	Significantly increases water-use efficiency in agriculture via targeted drip irrigation, minimizing waste.

Individual Member Contribution

Member	Contribution or responsibility
Purvikka N	Team Leader, Collecting Research Papers, Chart works, Data Collection
Kaviyen R	PPT editing, Data Collection, Research Papers
Ashwanth R	PPT editing, chart work, Data Collection
Akash Parthiban	Chart Work, Data Collection, Collecting Journal papers

Yukti ID

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